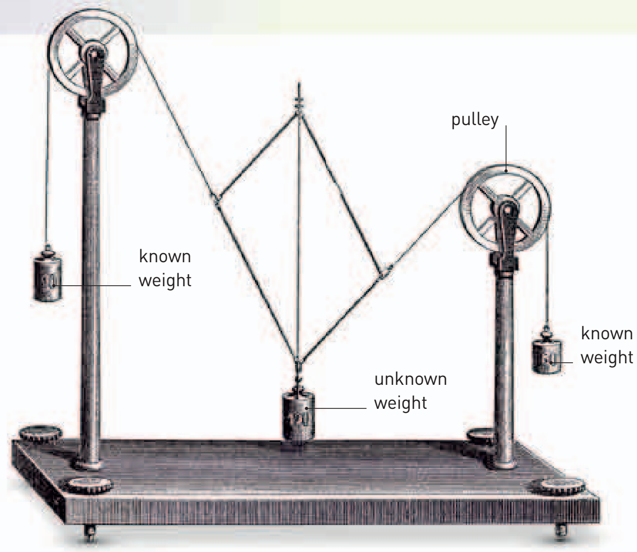
> **“** THE **LAW OF GRAVITATION** IS RENDERED **PROBABLE,** THAT **EVERY PARTICLE ATTRACTS** EVERY OTHER PARTICLE WITH **A FORCE WHICH VARIES INVERSELY** AS THE SQUARE OF THE DISTANCE. **”**

Isaac Newton, English mathematician, from *Principia,* 1687



First law of motion
According to Newton, these weights stay still because no net force acts upon them. Unknown weight can be calculated if the forces acting to keep it still are known.

Newton described the laws of motion and universal gravitation that became the foundation of physical science. The appearance of *Principia* was due in part to the efforts of Edmond Halley. At a time when the Royal Society had already spent its annual publishing budget, Halley stepped in to finance its production. He had even been responsible for Newton starting work on it in the first place.

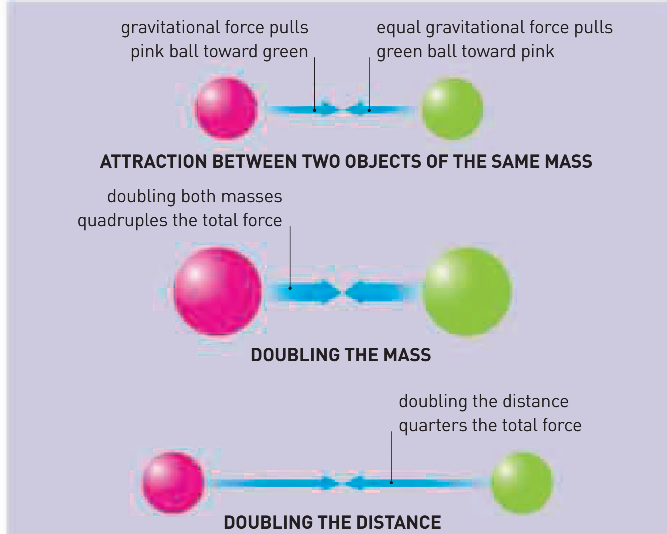
Three years earlier, three members of the Royal Society—**Christopher Wren, Robert Hooke**, and **Halley**—were debating **mathematical laws that govern the orbits of planets**, and Halley asked Newton for help in resolving a technical matter. Newton's response was a manuscript on planetary motion; impressed, Halley asked Newton to prepare a more exhaustive text for the Royal Society. For more than a year, Newton immersed himself in a study of physical laws, the result of which was his three-part masterpiece. In *Principia*, Newton describes his **three laws of motion** (see pp.120–21) and the **universal law of gravitation** (see panel, right), the basis of the branch of physics dealing with forces and motion: mechanics.

32.2
NEWTONS
THE **GRAVITATIONAL FORCE** PULLING ON **1 LB MASS** ON EARTH

Just before his death, Polish astronomer **Johannes Hevelius** (1611–87) completed the most **comprehensive celestial atlas** and star catalog of the time, in which he identified several new constellations, including Triangulum Minus. His work was published a few years later. In 1688, German astronomer **Gottfried Kirch** (1639–1710), director of the Berlin Observatory, described **another new constellation**, named Sceptrum Brandenburgicum in honor of the royal Prussian province. Today, its stars are considered part of the constellation Eridanus.

Naturalists continued to chart the diversity of the living world. In France, the botanist **Pierre Magnol** (1638–1715) had just become curator of France's biggest botanical garden at Montpellier. Magnol corresponded with English naturalist **John Ray**, who had embarked on his own survey of plant groups. Both men followed the principle of **classifying species according to anatomical similarities**. Their work implied underlying affinities within plant groups, although the evolutionary implications were not fully recognized for nearly two centuries. Magnol published his work in 1689 and, remarkably, many of his plant families are still recognized.

One of the earliest books on **pediatric medicine** appeared in 1689, published by English physician Walter Harris (1647–1732). This treatise on the diseases of children became a standard text on the subject.



UNIVERSAL LAW

Newton applied the physics of planetary interactions to create a Universal Law of Gravitation. Gravity is the force of attraction between bodies: stronger for more massive objects, weaker for a bigger distance apart. But whereas force and mass have a simple relationship, that between force and distance follows an inverse-square rule—doubling the distance reduces force by a quarter.

1687 Johannes Hevelius completes his **star atlas** shortly before his death.

1688 Gottfried Kirch describes the **new constellation** Sceptrum Brandenburgicum

1689 Pierre Magnon publishes his **classification of plants**, identifying families

1689 Walter Harris publishes one of the **earliest works on pediatric medicine**